Introduction to Theoretical Computer Science

CS2490 - Session 20: Pumping Lemma (Cont'd)

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Recall

In proving that a language is not regular, we are playing a game against an opponent (the universe):

- Opponent tries to trick us by choosing a pumping length p. (The number of states of a DFA supposedly recognizing L.)
- We respond with a word $w \in L$ adapted to the opponent's move (that is, $|w| \geq p$).
- Opponent tries to counter our move by choosing a tricky factorization $w = xyz$ (with $|y| \geq 1$ and $|xy| \leq p$). (The location of the loop in the computation of the DFA on w.)
- We administer the *coup de grâce* by finding an i such that $xy^i z \notin L$.

More precisely, we design a winning strategy to beat the opponent's moves, whatever they may be.

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Question: Our reasoning assumes $p \geq 2$. Can we do that?

Example 4 (cont'd)

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Remember that the p given by the opponent is the number of states of a DFA supposedly recognizing L. If our opponent chooses $p = 1$, then we can easily disprove that their DFA recognizes L since the only DFA with 1 states recognize either $\emptyset$ or Σ^*.

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Question: Can we choose n, m, ℓ and i so that we can complete the template?

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Spoiler: YES!

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CAREFUL! We need to choose n, m, ℓ such that for any choice $k \geq 1$ (from the sneaky universe), there exists an $i \geq 0$ such that $\ell = n + m + k(i-1)$. In particular, n, m and ℓ cannot depend on k, since they must be chosen before k is known.

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Choose for example $n = p$, $m = 1$ and $\ell = p! + p + 1$. Then, for any $1 \leq k \leq p$, we can choose $i = \frac{\ell - n - m}{k} + 1 = \frac{p!}{k} + 1 = 1 \times 2 \times \cdots \times (k-1) \times (k+1) \times \cdots \times p + 1$.

Example 4 (conclusion)

Prove that the language $L = \{0^n 1^m 0^\ell \mid n, m, \ell \geq 0 \wedge \ell \neq n + m\}$ is not regular.

Assume it is. Let p be the pumping length given by the Pumping Lemma.

We choose the word $w = 0^p 1^1 0^{p!+p+1}$. Note that $w \in L$ (suppose $p \geq 2$) and $|w| = 2p + p! + 1 \geq p$.

Let $w = xyz$ be any factorization such that $|y| \geq 1$ and $|xy| \leq p$. Since the first p letters of w are all 0, we have $y = 0^k$ for some $k \geq 1$.

Pump $i = \frac{p!}{k} + 1$ times! Indeed, $xy^i z = 0^{p+k(i-1)} 10^{p!+p+1} = 0^{p+p!} 10^{p!+p+1}$ is supposedly in L. But it clearly violate the condition to be in the language, so $xy^i z \notin L$.

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$$\{0^n1^m0^{n+m} \mid n, m \geq 0\} = (\Sigma^* \setminus L) \cap \mathrm{L}(0^*1^*0^*)$$

So, if L were regular, then $\Sigma^* \setminus L$ would be regular (complement of a regular language). Since $\mathrm{L}(0^*1^*0^*)$ is regular, then $\{0^n1^m0^{n+m} \mid n, m \geq 0\}$ would be regular (intersection of two regular languages). But we know it is not! Thus, L cannot be regular.

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- Use the Pumping Lemma to prove that a language is not regular.
- Always use the same template as shown in the examples.
- When choosing w is not straightforward, try to make a template and complete it once you found all the conditions your template must satisfy.
- You might have a plethora of way to complete the template. That's fine!
- If the condition to be in the language is *negative*, maybe it's easier to use the Pumping Lemma on the *positive* variation of the language. Then, use closure properties of regular languages to conclude.